Equivariant and N_∞-operads

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Main Take-Aways

Equivariantize All The Things!

Allow the group to act on any parameterizing diagrams.

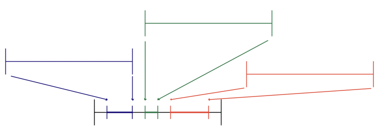
Internal & External Forms

Equivariant structures often have two equivalent forms:

$\{$Mackey, Tambara, etc$\}$ versus G-commutative monoids.

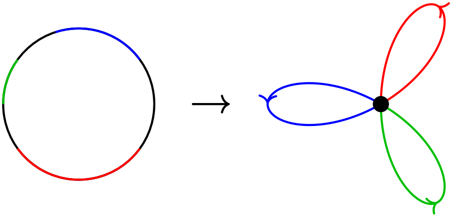
Little intervals

Embedding Intervals



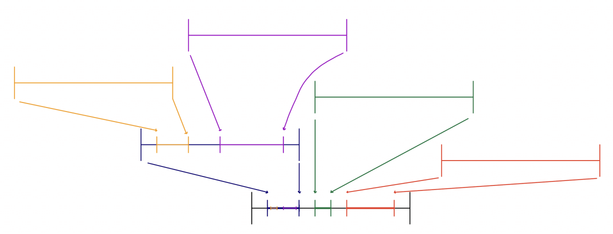
Look at the space

$$\mathcal{E}_1(n) = \mathrm{Emb}\left(\coprod^{n} I, I\right).$$



Composition

Associativity



$$\mathcal{E}_1(3) \times \big(\mathcal{E}_1(2) \times \mathcal{E}_1(1) \times \mathcal{E}_1(1)\big) \to \mathcal{E}_1(4).$$

Permuting Coordinates

Have a Σ_n action on $\mathcal{E}_1(n)$

Operads & G-Operads

Definition (May)

An operad is

1. *a sequence of* $G \times Sigma_n$*-spaces* $\mathcal{O}(n)$
2. *an "identity"* $i \in \mathcal{O}(1)$*, and*
3. *composition maps*

$$\mathcal{O}(n) \times \mathcal{O}(k_1) \times \cdots \times \mathcal{O}(k_n) \to \mathcal{O}(k_1 + \cdots + k_n).$$

The identity is an identity for the composition, and the compositions associate and respect reordering the factors.

Operadic Algebras

Definition

An $\mathcal{O}$-algebra is a space X with maps $\mathcal{O}(n) \times X^{\times n} \to X$ compatible with composition.

Example

The little disks in an orthogonal G-representation V:

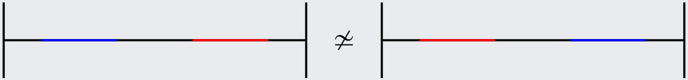
$$\mathcal{D}(V)_n = \mathrm{Emb}\left(\coprod^{n} D(V), V\right)$$

with the identity in $\mathcal{D}(V)_1$ as the identity, and with gluing in embeddings as composition.

Higher commutativity

Little intervals

The space $\mathcal{E}_1(2)$ is not path connected:



Little cubes

As k grows, the spaces

$$\mathcal{D}(\mathbb{R}^k)(n) = \mathrm{Emb}\left(\coprod^{n} D^k, D^k\right)$$

become more and more connected.

Coherent Commutativity

$\mathcal{E}_\infty$

An $\mathcal{E}_\infty$-operad has $\mathcal{E}_\infty(n)$ a contractible, Σ_n-free space.

What does this give us?

Any choice of Σ_n-equivariant map $\Sigma_n \to \mathcal{E}_\infty(n)$ gives a multiplication

$$X^n \cong \Sigma_n \underset{\Sigma_n}{\times} X^n \to \mathcal{E}_\infty(n) \underset{\Sigma_n}{\times} X^n \to X.$$

Slogan

Ordinary $\mathcal{E}_\infty$-operads structure “coherently commutative” ordinary multiplications, and there is only one, up to homotopy.

Non-trivial actions?

G-operads

What changes do non-trivial G-actions introduce?

Equivariant Little Disks

What happens for non-trivial representations?

Higher Commutativity

What plays the role of $\mathcal{E}_\infty$ here?

Elmendrof's Theorem Homotopy

Coefficient Systems of Spaces

Given a G-space X, we have a functor $\underline{X}\colon \mathcal{O}rb^{G,op} \to \mathcal{T}op$:

$$T \mapsto \mathrm{Map}^G(T, X).$$

Theorem (Elmendorf)

The assignment $X \mapsto \underline{X}$ gives an equivalence of infinity categories

$$\mathcal{T}op^G \simeq \mathrm{Fun}(\mathcal{O}rb^{G,op}, \mathcal{T}op).$$

Rewriting $\mathcal{D}(V)$

Equivariant Homotopy Type

Evaluation at the origins gives an equivalence

$$\mathcal{D}(V)(n) \xrightarrow{\simeq} \mathrm{Emb}\big(\{1,\dots,n\}, D(V)\big) \cong \mathit{Conf}_n(V).$$

Action

An element $(g,\varsigma) \in G \times \Sigma_n$ acts on an embedding $f\colon \{1,\dots,n\} \to V$ as

$$(g,\varsigma)\cdot f(j) = g\Big(f\big(\varsigma^{-1}(j)\big)\Big).$$

Fixed points of $\mathcal{D}(V)$

Proposition

Σ_n acts freely on $\mathcal{D}(V)(n)$.

Proposition

If $\Gamma \subset G \times \Sigma_n$ has $\Gamma \cap \Sigma_n = \{e\}$, then

$$\Gamma = \Big\{ \big(h, f(h)\big) \mid h \in H \subseteq G, f \colon H \to \Sigma_n \Big\}.$$

Graph Subgroups

Graph Subgroups of $G \times \Sigma_n$ are the same data as a subgroup H and an H-set structure on $\{1, \ldots, n\}$.

Fixed points of $\mathcal{D}(V)$

Theorem

Let Γ be a graph subgroup corresponding to a finite H-set T of cardinality n. Then

$$Conf_n(V)^\Gamma \cong \mathrm{Emb}^G(T, V).$$

Corollary

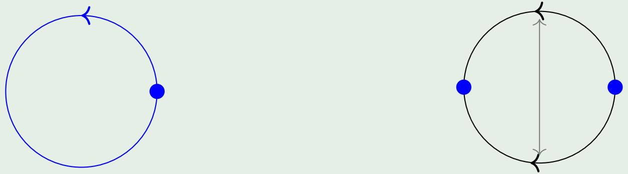
Let X be a $\mathcal{D}(V)$-algebra. Then we have $\mathrm{Emb}^G(T, V)$-maps

$$\mathrm{Map}(T, X) \to X.$$

C_2-equivariant 1-spheres

Example

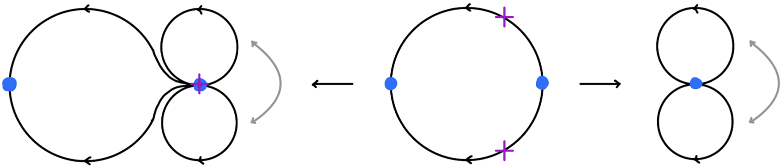
If $G = C_2$, then we have S^1 and S^σ, the unit complex numbers with complex conjugation.



Two kinds of loop spaces

Have $\Omega X = \mathrm{Map}_*(S^1, X)$ and $\Omega^\sigma X = \mathrm{Map}_*(S^\sigma, X)$.

Signed loop spaces



"Multiplications"

$$\Omega^\sigma X \times \left(\Omega X \underset{\leftrightarrows}{\times} \Omega X\right) \to \Omega^\sigma X \leftarrow \left(\Omega X \underset{\leftrightarrows}{\times} \Omega X\right).$$

No ordinary product

$\Omega^\sigma X$ does not have a multiplication! $\Omega^\sigma X \times \Omega^\sigma X \not\to \Omega^\sigma X$

Structure on signed loops

Underlying

- Non-equivariantly ΩX is an ordinary loop-space
- C_2 acts as an anti-automorphism: the inverse

Fixed Points

- Have an action of ΩX on $\left(\Omega^\sigma X\right)^{C_2}$, and
- Have a "unit" map $\Omega X \to \left(\Omega^\sigma X\right)^{C_2}$.

Mackey Functors

The signed-loop action gives us a transfer map!

N_∞ Operads

Definition (Blumberg–H.)

An N_∞ operad for G is a G-operad $\mathcal{O}(n)$ such that

1. *Σ_n-acts freely,*
2. *if $\Gamma \subseteq G \times \Sigma_n$, the fixed points $\mathcal{O}(n)^\Gamma$ are either empty or contractible, and*
3. *the G-fixed points $\mathcal{O}(n)^G$ are always contractible.*

Example

If U is a universe, then

1. $\mathcal{D}(U)$ is N_∞
2. the linear isometries operad $\mathcal{L}(U)$ is N_∞

Rewriting: Admissible Sets

Definition

A finite H-set T is admissible for $\mathcal{O}$ if $\mathcal{O}(|T|)^{\Gamma_T} \neq \emptyset$ for Γ_T any graph subgroup classifying the H-set T.

Example

The finite G-set $* \amalg \cdots \amalg *$ is admissible for any $\mathcal{O}$.

Example

T is admissible for $\mathcal{D}(U)$ iff $T \hookrightarrow U$.

Example

T is admissible for $\mathcal{L}(U)$ iff $\mathbb{R} \cdot T \otimes U \hookrightarrow U$.

N_∞ operads do the right thing

Theorem (Blumberg–H.)

If T is an admissible H-set for $\mathcal{O}$, then $\mathcal{O}$ structures an essentially unique H-equivariant twisted multiplication

$$\mathrm{Map}(T, X) \to X.$$

Example

Little disks for $\infty\rho$. We have all transfers

$$\mathrm{Map}(T, X) \to X.$$

Combinatorial implications

Interconnected Transfers

T being admissible for H implies the existence of lots of other admissible sets

Example

The transfer

$$\mathrm{Map}(C_2, \Omega X) \to \Omega^\sigma X$$

gave the ordinary loop multiplication.

Theorem

The admissible sets are a complete invariant of N_∞ operads.

Indexing Systems

Definition (Blumberg–H.)

An indexing system is a collection of subcategories $\underline{\mathcal{O}}(H) \subseteq \mathcal{F}in^H$, one for each subgroup $H \subseteq G$ such that

1. *$\underline{\mathcal{O}}(H)$ is closed under disjoint unions and summands,*
2. *if $K \subseteq H$, then $T \in \underline{\mathcal{O}}(H)$ implies $i_K^* T \in \underline{\mathcal{O}}(K)$,*
3. *if $g \in G$, then $T \in \underline{\mathcal{O}}(H)$ implies $c_g^* T \in \underline{\mathcal{O}}(gHg^{-1})$,*
4. *if $H/K \in \underline{\mathcal{O}}(H)$, then $T \in \underline{\mathcal{O}}(K)$ implies $H \underset{K}{\times} (T) \in \underline{\mathcal{O}}(H)$.*

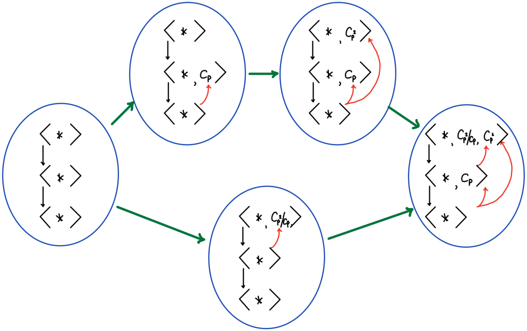
Poset

The admissible H-sets as H varies forms an indexing system.

Poset of Indexing Systems

The Poset $\mathcal{I}(G)$

$\underline{\mathcal{O}} \leqslant \underline{\mathcal{O}}'$ if for all H, $\underline{\mathcal{O}}(H) \subseteq \underline{\mathcal{O}}'(H)$.



Transfer Systems

Definition (Rubin, Balchin–Barnes–Roitzheim)

A transfer system on G is a poset structure $\to$ on $Sub(G)$ satisfying

1. *If $K \to H$, then $K \subset H$,*
2. *if $K \to H$ and $g \in G$, then $gKg^{-1} \to gHg^{-1}$, and*
3. *if $K \to H$ and $J \subset H$, then $K \cap J \to J$.*

Theorem (Rubin, Balchin–Barnes–Roitzheim)

$$Tran(G) \cong \mathcal{I}(G).$$

Classification Theorem

Theorem (Blumberg–H.)

$\mathcal{O} \mapsto \underline{\mathcal{O}}$ gives a fully faithful embedding of ∞-categories

$$\mathcal{N}_\infty^G \hookrightarrow \mathcal{I}(G).$$

Theorem (Bonventre–Pereira, Gutiérrez–White, Rubin)

Every indexing system is the indexing system associated to an N_∞-operad.

Equivariant Algebra

Rewriting Coefficient Systems

$$\mathcal{C}oeff \simeq \mathrm{Fun}^{\Pi}\left(\mathcal{F}in^{G,op}, \mathcal{S}et\right)$$

Coefficient Systems of Sets

If X is pointed,

$$\underline{\pi}_k(X)(T) := \pi_k\Big(\mathrm{Map}(T, X)\Big).$$

Contravariance

Definition

If T is a finite G-set and $\underline{M}$ is a coefficient system, let $\underline{M}_T$ be the coefficient system

$$S \mapsto \underline{M}(T \times S).$$

Proposition

This gives a diagram $\mathcal{F}in^{G,op} \to \mathcal{C}oeff^G$.

Extending Composites

Have a natural isomorphism of functors

$$\underline{M}_{G/H} \cong \mathrm{CoInd}_H^G i_H^* \underline{M}.$$

Coefficient Systems of Abelian Groups

Definition

A coefficient system of abelian groups is a product preserving functor

$$\mathcal{F}in^{G,op} \to \mathcal{A}b.$$

Adjunctions

If $H \subseteq G$, then we have a forgetful functor $i_H^*\colon \mathcal{C}oeff^G \to \mathcal{C}oeff^H$ given by

$$i_H^* \underline{M}(T) := \underline{M}(G \underset{H}{\times} T).$$

This has right adjoint

$$\mathrm{CoInd}_H^G \underline{M}(T) := \underline{M}(i_H^* T).$$

Example: C_p-Coefficient Systems

$$
\begin{tikzcd}
0 \arrow[r] \arrow[d] & \underline{M}(C_p/C_p) \arrow[r, "res"] \arrow[d] & \underline{M}(C_p/e) \arrow[d] \\
\mathbb{Z}[G] \otimes \underline{M}(C_p/e) \arrow[r] & \underline{M}(C_p/e) \arrow[r] & \mathrm{Hom}(\mathbb{Z}[G], \underline{M}(C_p/e))
\end{tikzcd}
$$

$$\mathrm{Ind}_e^{C_p}(i_e^*\underline{M}) \qquad \underline{M} \qquad \mathrm{CoInd}_e^{C_p} i_e^*\underline{M}$$

$\mathcal{D}(V)$-algebras

Transfers Give Transfers

If T is an admissible set for $\mathcal{O}$, then $\underline{\pi}_k(X)$ has a T-transfer:

$$\underline{\pi}_k(X)_T \to \underline{\pi}_k(X).$$

Example

This gives us a map

$$\underline{\pi}_k(X)(C_p/e) \to \underline{\pi}_k(X)(C_p/C_p)$$

that

1. factors through the coinvariants
2. restricts to the trace.

Mackey Functors

Proposition

If X is a $\mathcal{D}(\infty\rho)$-algebra in spaces, then for all k, the functor on $\mathcal{F}in^{G,\cong}$

$$T \mapsto \underline{\pi}_k(X)_T$$

assemble to a covariant functor

$$\mathcal{F}in^G \to \mathcal{C}oeff.$$

Mackey Functors

This is the *external* form of a Mackey functor.